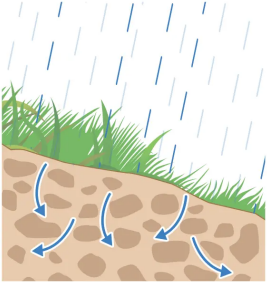
Evaluation of ECMWF S2S Forecasts over Ireland for Multi-Hazard Applications

Lainey Ward

Decarb-AI Centre, University College Dublin

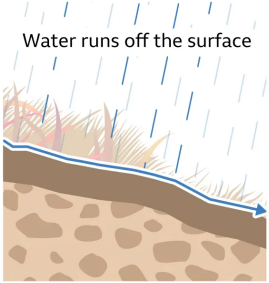
What Happens When Drought Becomes Flood?

Normal conditions



Soil absorbs water like a sponge

Drought conditions



Hard layer of soil repels water

Adapted from BBC Research (2022)

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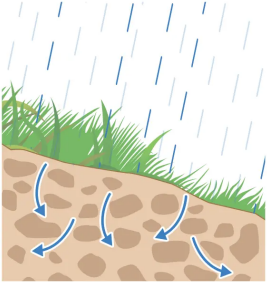

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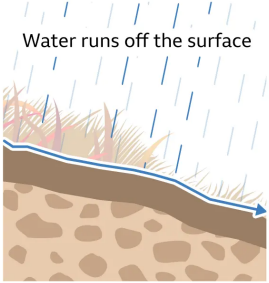

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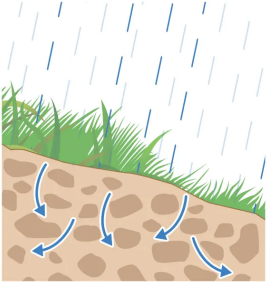

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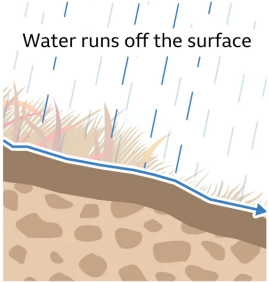

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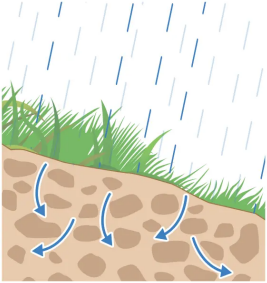

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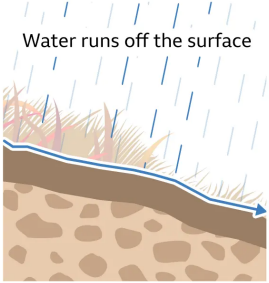

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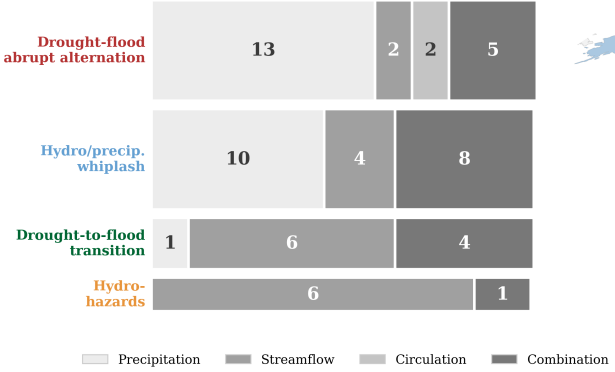
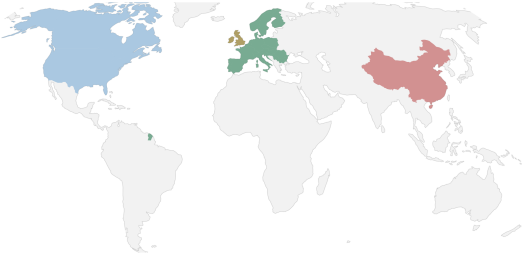
compound events by 2050

Steensen et al. (2025)

What Does the Research Look Like?





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Parry et al. (2023)

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Propagation to catchment level

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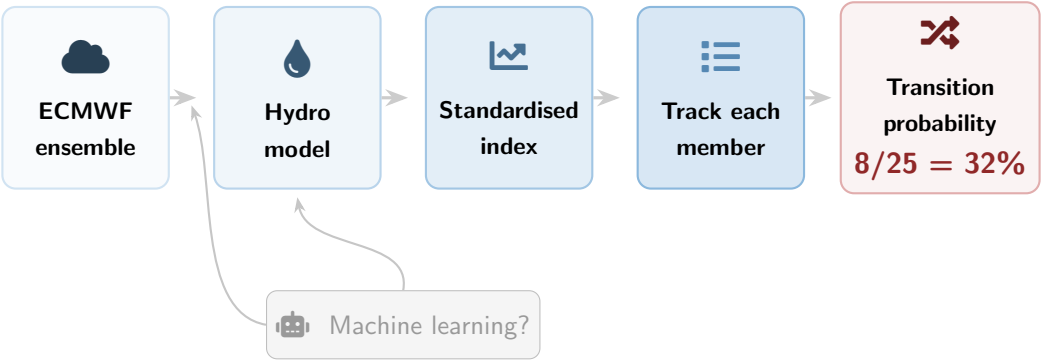
Open questions

?!

Skill?

Define transition?

What's My Approach?



Takeaways

Same phenomenon, four distinct traditions

Nobody has forecast them yet

Ireland as my testbed

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References

Brunner, M.I., Anderson, B., Muñoz-Castro, E. (2025). Meteorological and hydrological dry-to-wet transition events are only weakly related over European catchments. *Environ. Res. Lett.*, 20(8), 084013.

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